



Hoisting

HOISTING SAFETY INFORMATION

The information contained on this page and the following page is critical to the safe application and use of hoisting components. Please read and understand all of the information contained in these pages **before** designing, building, or operating any equipment or device incorporating any of these hoisting components. Designs incorporating these hoisting products and the subsequent use of them is intended to be only by trained, qualified, and experienced persons. Misapplication, misuse, or lack of supervision and inspection can lead to serious accidents including death. Prior to the use of any of these products, evaluation of the product application, safety precautions, safe working load, and control of all field conditions is mandatory. Be cautioned that all safety factors are approximate. **Safe working load should never be exceeded under any circumstances.**

If you should have any questions about the proper use or installation of any of our products, please be free to contact our ready technical support.

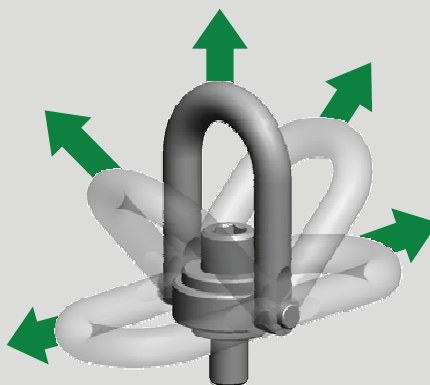
- 1) Despite the 5:1 safety factor on hoist rings, never exceed the rated load capacity. This safety margin is needed in case of misuse which could drastically reduce load capacity.
- 2) Tensile strength of parent material should be above 80,000 psi to achieve full load rating. For weaker material, consider through hole mounting with a nut and washer on the other side.
- 3) Do not allow hoist rings to bind. Use a spreader bar if necessary.
- 4) Do not use spacers, shims, or washers between the hoist ring and the mounting surface.
- 5) The mounting surface must be flat and smooth for full contact under hoist ring. Tapped mounting holes must be perpendicular to the mounting surface.
- 6) Tighten mounting screws to the torque recommended. Periodically check torque because screws could loosen in extended service. For hoist rings not furnished with screws, mount using only high-quality socket head cap screws.
- 7) Never lift with a hook or other device that could deform the hoist ring.
- 8) Do not apply shock loads - always lift gradually. Repeat magnaflux testing if shock loading ever occurs.
- 9) After installation, check that the ring rotates and pivots freely in all directions.
- 10) Never use a hoist ring that shows signs of wear, cracks, or corrosion. Any of these factors will significantly reduce load rating and may limit the hoist ring from rotating and pivoting freely.

HOIST RINGS vs. EYE BOLTS:

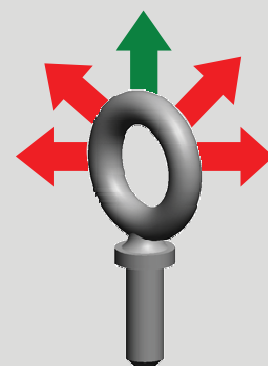
Eye bolts have been around for over a hundred years and have been doing an excellent job when used properly. The proper use of an eye bolt means that the load direction is 0 degrees. If the load shifts or the direction of loading on the eyebolt is as much as 10 degrees off the zero line, the result is a bent eye bolt. A bent eye bolt is a broken eye bolt, and will result in failure if the eye bolt is used any longer.

Swivel hoist rings are designed for lifting at any angle because they can swivel 180 degrees. Therefore, the rated load remains the same, regardless of the angle of the force.

Same load rating at any angle.



Only safe at zero degrees.





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IMPORTANT:

The load on each hoist ring is not simply total weight divided by the number of hoist rings. The resultant force can be significantly greater at shallow lift angles and with unevenly distributed loads. See the examples and chart below.

L = Load experienced by hoist ring

N = Number of hoist rings = 4

W = Total weight = 2,000 lbs

A = Lifting angle

$$L = \frac{W}{N \sin A}$$

EXAMPLE:

A 2,000 lbs plate is being lifted by 4 hoist rings with a lifting angle of 60°.

$$L = \frac{2000}{4 \sin 60} = 577 \text{ lbs}$$

In this example each of the four hoist rings will experience 577 pounds.

A 2,000 lbs plate is being lifted by 4 hoist rings with a lifting angle of 20°.

$$L = \frac{2000}{4 \sin 20} = 1,462 \text{ lbs}$$

Now in this example each of the four hoist rings will experience 1,462 pounds because of the reduced lifting angle.

LIFTING ANGLE	NUMBER OF HOIST RINGS	WEIGHT OF LOAD (POUNDS)	ACTUAL APPLIED LOAD ON HOIST RING (LBS)
90°	4	2,000	500
80°	4	2,000	510
70°	4	2,000	535
60°	4	2,000	580
50°	4	2,000	655
40°	4	2,000	780
30°	4	2,000	1,000
20°	4	2,000	1,465

PROPER LIFTING ANGLE:

Proper lifting angle is extremely critical in the safe operation of hoist rings. We ask that you pay special attention to the information above and fully understand all facets of these components before dealing with them in any way. If you are ever unsure about the installation or operation of any hoisting component please do not hesitate to call our technical support personnel.

